

James Oswald

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Research Interests

Artificial Intelligence	Automated Reasoning, Cognitive Architectures, Machine Intelligence Measures
Machine Learning	Generative Models, Reinforcement Learning, Large Language Models
Logic	Modal Logics, Description Logics, PDL

Education

Rensselaer Polytechnic Institute

Troy, NY

PHD IN COMPUTER SCIENCE (PURSUING)

Aug. 2021 - Present

- Concentration in Artificial Intelligence
- Sub-concentrations in in Theory of Computation and Systems/Software
- Advised by Prof. Selmer Bringsjord and Prof. Thomas Ferguson
- Graduate Researcher in the Rensselaer Artificial Intelligence and Reasoning (RAIR) Lab
- 4.0 GPA

The State University of New York at Albany (SUNY Albany)

Albany, NY

B.S. IN COMPUTER SCIENCE AND MATHEMATICS

Aug. 2018 - May. 2021

- Double Major in Computer Science and Mathematics
- Undergraduate Researcher in the Data Mining and Management (DMM) Lab
- Graduated *summa cum laude* (3.89 GPA)

Research Employment

United States Naval Research Laboratory (NRL)

Washington, DC

SSEP STUDENT

July. 2026 - Present

- Integrated Cognitive Systems Section at the Navy Center For Applied Research In Artificial Intelligence.

CATS Laboratory @ RPI

Troy, NY

GRADUATE RESEARCH ASSISTANT

Nov. 2025 - Present

- CRAFT (Center for Research toward Advancing Financial Technologies) grant
"A Neurosymbolic Semantic Layer for compliance checking."

Rensselaer AI and Reasoning Laboratory @ RPI

Troy, NY

GRADUATE RESEARCH ASSISTANT

Jan. 2025 - Jan. 2026

- Applied for and was awarded a \$120,000 Rensselaer-IBM FCRC grant:
"Large Language Models as Planning Domain Model Generators."

United States Naval Research Laboratory (NRL)

Washington, DC

SSEP STUDENT

June. 2025 - Sep. 2025

- Integrated Cognitive Systems Section at the Navy Center For Applied Research In Artificial Intelligence.

United States Naval Research Laboratory (NRL)

Washington, DC

SSEP STUDENT

July. 2024 - Sep. 2024

- Integrated Cognitive Systems Section at the Navy Center For Applied Research In Artificial Intelligence.

International Business Machines (IBM)

RENSSELAER-IBM AIRC RESEARCH EXTERN

Yorktown Heights, NY

May. 2023 - Aug. 2023

- Working under Kavitha Srinivas and the IBM AI Planning Research Group.

Rensselaer AI and Reasoning Laboratory @ RPI

GRADUATE RESEARCH ASSISTANT

Troy, NY

May - Sep. 2022, Jun. - Aug. 2024

- Funded by ONR Grant #N00014-22-1-2201 "Meta-Cognitively Perceptive Artificial Teammates"

The Research Foundation for SUNY

RESEARCH AIDE

Albany, NY

Jan. 2021 - Aug. 2021, Jan. 2022

- Funded By NSF Grant #2025793: "Generative Machine Learning for Nanomaterial Design"

Teaching and Mentorship Experience

RPI Graduate Teaching Assistant

Aug 2021 - Dec 2025

- Introduction to Logic Based AI (Fall 2025)
- Introduction to Formal Logic and AI (Spring 2025)
- Introduction to Logic Based AI (Fall 2024)
- Introduction to Formal Logic and AI (Spring 2024)
- Intermediate Formal Logic and AI (Fall 2023)
- Data Structures (Spring 2023)
- Computer Science I (Fall 2022)
- Data Structures (Spring 2022)
- Programming Languages (Fall 2021)

SUNY Albany Capstone Project Sponsor

Fall 2022 - Present

I propose and manage capstone projects for the final course in the UAlbany undergraduate computer science curriculum (ICSI 499).

- A Smart Matter Terminal (Spring 2026)
- A Cross-platform App for the Evaluation of Geospatial Reasoning Ability (Fall 2024)
- A Web Application for Modeling Peirce's Alpha Existential Graph System (Fall 2023)
- A 3D Virtual Reality Interface for Graphical Interactive Theorem Proving (Fall 2023)
- A Test Suite for the Evaluation of The NARS Cognitive Architecture on 3D Reasoning (Fall 2022)

RPI Center for Open Source (RCOS) Project Sponsor

Fall 2024

I propose and manage RCOS projects for RPI undergraduate students.

- An Application Modeling Peirce's Gamma Existential Graph System (Fall 2024)

Professional Service / Reviewing

PC Member AGI (2023, 2024, 2025, 2026), ICAPS (2025), ICML (2026)

Subreviewer IJCAI (2021), DSAA (2021)

Journal Reviewer J. Log. Lang. Inf. (2025)

Selected Publications

Key:

Forthcoming - accepted for publication but not published yet.

Under review - submitted but not yet accepted for publication.

COMPETITIVELY REFEREED CONFERENCE AND WORKSHOP PAPERS

1. Timofey Tylik, Kyle Fuller, **James Oswald**, Thomas Ferguson, Selmer Bringsjord. "A Legg-Hutter Intelligence Measure for Continuous-Time Analog Agents and Environments". Artificial General Intelligence (AGI), 2026.

2. **James Oswald**, Daniel Obolensky, Volodymyr Varha, Vasilije Dragovic, Kavitha Srinivas, Harsha Kokel, Michael Katz, Shirin Sohrabi. "Model Space Reasoning as Search in Feedback Space for Planning Domain Generation". ICLR Workshop on World Models, 2026.
3. Viane Matsibekker, Gwendolyn Morgan, **James Oswald**, Thomas M Ferguson. "A Logic-Driven Framework for Home Health Monitoring Using the Matter Protocol and Epistemic Reasoning." IEEE International Conference on Smart Internet of Things (SmartIoT), 2025/
4. **James Oswald**, Thomas Ferguson, Selmer Bringsjord. "On the Arrowian Impossibility of Machine Intelligence Measures". Artificial General Intelligence (AGI), 2025.
5. Kyle Fuller, Deacon Sawyer, **James Oswald**, Thomas Ferguson. "Resource Relativized Legg-Hutter Intelligence". Artificial General Intelligence (AGI), 2025.
6. **James Oswald**, Thomas Ferguson, Selmer Bringsjord. "A Universal Intelligence Measure for Arithmetical Uncomputable Environments". Artificial General Intelligence (AGI), 2024.
7. **James Oswald**, Kavitha Srinivas, Harsha Kokel, Junkyu Lee, Michael Katz, Shirin Sohrabi. "Large Language Models as Planning Domain Generators." International Conference on Automated Planning and Scheduling (ICAPS), 2024.
8. **James Oswald**, Brandon Rozek. "Parallel Verification of Natural Deduction Proof Graphs". Logical Frameworks and Meta-Languages: Theory and Practice (LMFTP), 2023.
9. Selmer Bringsjord, John Slowik, NS Govindarajulu, Michael Giancola, **James Oswald**, Rikhiya Ghosh, "Affect-based Planning for a Meta-Cognitive Robot Sculptor: First Steps". Affective Computing and Intelligent Interaction Conference (ACII), 2023.
10. Selmer Bringsjord, **James Oswald**, Michael Giancola, Brandon Rozek, NS Govindarajulu. "The M Cognitive Meta-Architecture as Touchstone for Standard Modeling of AGI-Level Minds". Artificial General Intelligence (AGI), 2023.
11. Selmer Bringsjord, NS Govindarajulu, John Slowik, **James Oswald**, Michael Giancola, John Angel, Shreya Banerjee, Aiden Flaherty. "PERI.2 Goes to Preschool and Beyond, in Search of AGI". Artificial General Intelligence (AGI), 2022
12. **James Oswald**. "Market Prediction as a Task for AGI Agents", Artificial General Intelligence (AGI), 2022
13. Fariha Moomtaheen, Matthew Killeen, **James Oswald**, Anna González-Rosell, Peter Mastracco, Alexander Gorovits, Stacy M Copp, Petko Bogdanov. "DNA-Stabilized Silver Nanocluster Design via Regularized Variational Autoencoders". ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2022.

JOURNAL ARTICLES

1. **James Oswald**, Brandon Rozek, Thomas M. Ferguson. "Modeling C0 Family Logics for Artificial Intelligence: Doxastic-Temporal Logics for Reasoning About Goals". Künstliche Intelligenz, 2024.
2. Selmer Bringsjord, Michael Giancola, NS Govindarajulu, **James Oswald**, John Slowik, Paul Bello, Micah Clark. "Argument-Based Inductive Logics, With Coverage of Compromised Perception". Frontiers in Artificial Intelligence, 2023.

BOOK CHAPTERS

1. Selmer Bringsjord, NS Govindarajulu, **James Oswald**. "Universal Cognitive Intelligence, from Cognitive Consciousness, and Lambda". Computational Approaches to Conscious Artificial Intelligence, 2023.

Selected Awards

Springer Prize for Best Paper at AGI24

Aug 2024

For the paper: *A Universal Intelligence Measure for Arithmetical Uncomputable Environments*

RPI Departmental Letter of Recognition from Boleslaw K. Szymanski

Jan 2023

For the project: *An Analysis of Motifs in Multilayer Brain Networks Across Sex and Age*

SUNY Albany Presidential Award for Undergraduate Research

Jun. 2021

For the project: *Intelligent Nanomaterial Design using Variational Autoencoders*

SUNY Albany Presidential Scholarship

Aug. 2018 - May. 2021

Professional Memberships & Societies

Association for Symbolic Logic (ASL)

July. 2026 - Present

- Student Member (July. 2026 - Present)

Association for the Advancement of Artificial Intelligence (AAAI)

July. 2023 - Present

- Student Member (July. 2023 - Present)

Association for Computing Machinery (ACM)

July. 2023 - Present

- Student Member (July. 2023 - Present)
- SIGLOG Member (August 2026 - Present)
- SIGAI Member (August 2026 - Present)

Artificial General Intelligence Society (AGIS)

August. 2022 - Present

- Senior Member (Aug. 2022 - Present)

Institute of Electrical and Electronics Engineers (IEEE)

Jan. 2019 - Present

- Graduate Student Member (Jan. 2022 - Present)
- Undergraduate Student Member (Jan. 2019 - Jan. 2022)
- Computer Society Graduate Member (Jan. 2022 - Present)

University Involvement & Honor Societies

SUNY Albany Hackathon Staff (Hack-a-Damien)

Oct. 2024 - Present

- Judge (2024, 2025, 2026)

RPI Computer Science Honors Society - Upsilon Pi Epsilon

Nov. 2022 - Present

- Full Member (Nov. 2022 - Present)
- Website Committee Member (Oct. 2023 - May 2024.)
- Programming Competition Committee Member (Oct. 2023 - May 2024)

SUNY Albany IEEE Student Chapter

Oct. 2018 - Present

- Member of the Board of Trustees (May. 2023 - Present)
- Alumni Advisor (Jun. 2021 - May. 2023)
- Chair (May. 2020 - Jun. 2021)
- Junior Chair (Feb. 2020 - May. 2020)
- Member (Oct. 2018 - Feb. 2020)

NY Capital Region Logic Reading Group

May 2022 - Dec 2025

- Co-Founder and Organizer

SUNY Albany KDD Cup (Machine Learning Competition) Team

May 2021

- Team Captain (Apr. May 2021)

SUNY Albany Presidential Honors Society

Jan 2019 - May 2021

- Full Member (Jan 2019 - May 2021)